

1-A:

Public goods must be **non-rival** (one person's use doesn't reduce another's) and **non-excludable** (payment by one person should not restrict others from enjoying the benefits associated with the good; people can't be stopped from using it).

- **Education is excludable** (you can be denied access if you can't pay for it) and **rival** (classroom space is limited).
So, it is **not** a pure public good.

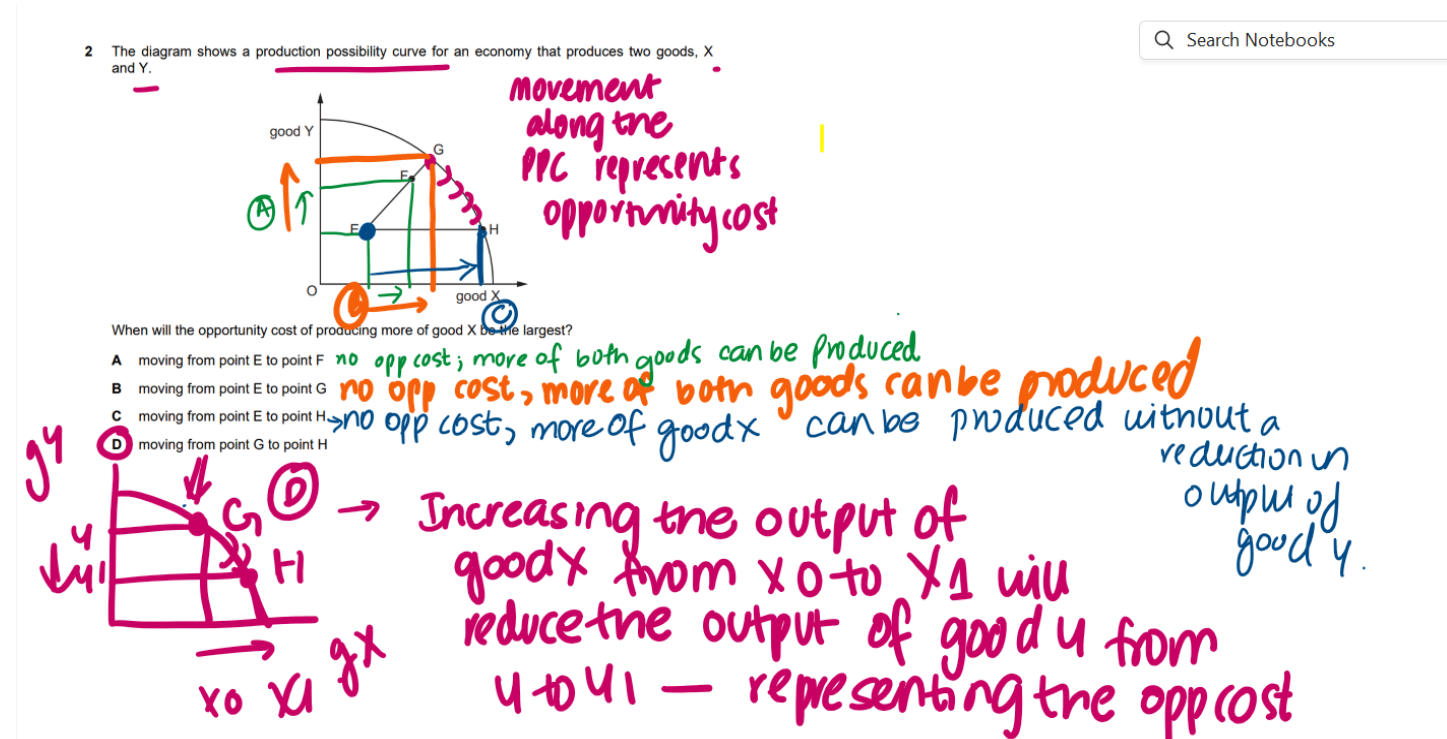
B: Flood control systems are non-rival and non-excludable — classic public goods.

C: National defence protects everyone, and is non-rival and non-excludable— definitely a public good.

D: Street lighting benefits all nearby — one user doesn't reduce others' use, and you can't easily exclude people. Hence, it is a public good.

2-D: **Opportunity cost** represents the value of the next best alternative that is forgone when a decision is made. Essentially, it's what you give up to get something else.

A movement along the PPC represents opportunity cost.



In the other options, we are already operating inside of the PPC, so increasing output won't incur opportunity cost.

3-C:

A **free good** is **abundant**, has **zero opportunity cost**, and no price. Fresh air in the countryside fits perfectly as it's available without reducing anyone else's supply; the situation of scarcity will not occur.

A: River water used by a farmer can cause scarcity downstream — not a free good.

B: Medical services from a charity have **opportunity costs** (staff, medicine) — not free in economic terms.

D: A gift may be **free to you**, but not to the giver; there was a cost involved.

4-A:

Scarcity exists because resources are **limited** while wants are **unlimited**. If we had **enough resources to meet all needs and wants**, scarcity would be eliminated. Hence, A is the correct answer.

B: Overproduction leads to waste, not necessarily elimination of scarcity.

C: A surplus budget is a **fiscal** issue, unrelated to overall scarcity.

D: Market equilibrium doesn't eliminate scarcity; it balances demand and supply at a price, but wants may still exceed what's available.

5-C:

Division of labour boosts productivity because workers specialise in tasks they're best at, reducing time wasted on switching and training. Firms aim to increase efficiency and output — this is the clearest motivation.

A: Self-sufficiency reduces specialisation, which is the opposite of division of labour.

B: Flexibility usually means multi-skilled workers, not specialised ones.

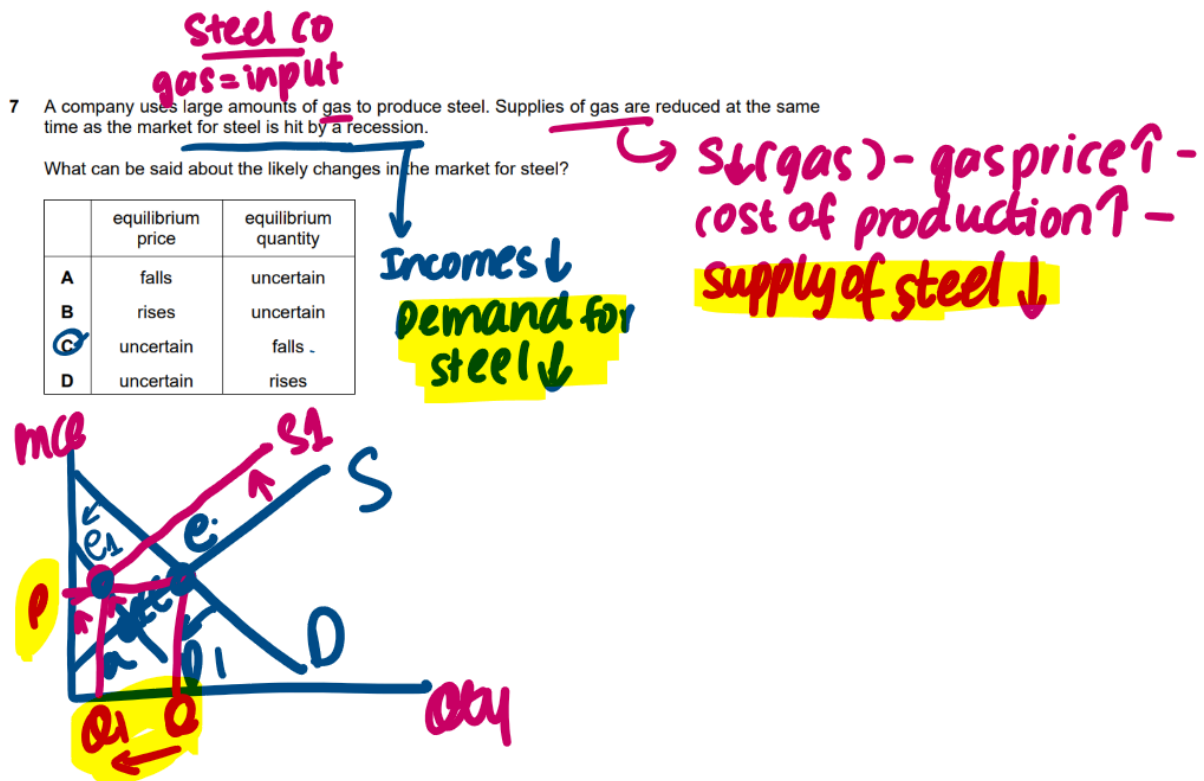
D: Lower risk is not the main driver for introducing division of labour.

6-D:

Oil exploration involves massive capital investment, long project timelines, regulatory approvals, and geophysical constraints. This makes the **short-run supply of oil exploration highly inelastic**, so output cannot quickly respond to increased demand. This means that even when demand for oil increases, which raises prices, oil explorers can't respond by extending supply significantly, as it takes time to explore oil.

A, B, C are incorrect as all (fast food restaurants, household cleaning services and newspaper printing) are **labour-intensive, service-based sectors** with relatively **low fixed costs** and **flexible input adjustment**. They can respond more quickly by hiring staff, working overtime, or adjusting order volumes. In other words, their **supply will be relatively elastic**.

7-C:



The reduction in gas supply pushes gas prices upwards, and increases production costs of steel, shifting the supply curve for steel leftward, while falling demand due to recession shifts the demand curve leftward. Both shifts reduce quantity, but the price effect is ambiguous, depending on the magnitude of shifts — hence uncertain.

8-A:

A firm is currently charging **\$12 per unit** and operating at **80% of its full production capacity**, which amounts to selling **8,000 units** out of a maximum of 10,000 units per month.

The question states that demand for the product is unitary price elastic, meaning that any percentage change in quantity demanded is exactly matched by an opposite percentage change in price.

To fully utilise its production capacity, the firm needs to increase output from **8,000 to 10,000** units, which is a **25% increase in quantity**.

Under unitary elasticity, **this 25% rise in quantity must be offset by a 25% fall in price**. Applying a **25% reduction to the original price of \$12 gives a new price of \$9.00**. Therefore, the firm should charge \$9.00 in order to sell 10,000 units and make full use of its capacity, making option A the correct answer.

Using the Price Elasticity of Demand Formula

$$PED = \frac{\% \Delta Q}{\% \Delta P}$$

Step	Description	Working	Result
1	Calculate % change in quantity	$\frac{10,000 - 8,000}{8,000} \times 100$	+25%
2	Apply PED = -1 (unitary elasticity)	$\% \Delta P = \frac{\% \Delta Q}{PED} = \frac{25\%}{-1}$	-25%
3	Calculate new price	$12 - (25\% \times 12) = 12 - 3$	\$9.00

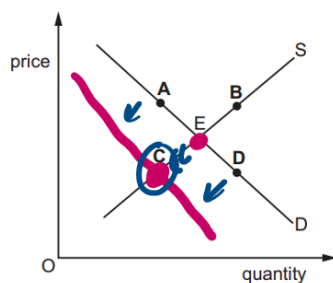
9-C:

4

- 9 The diagram shows the demand for and supply of hotel accommodation. The market equilibrium is at point E.

The hotel receives a very large number of bad reviews about the quality of its accommodation.

Which point on the diagram would show the new market equilibrium?



Negative reviews reduce **consumer preferences** for the hotel, shifting **demand to the left**. This creates a new equilibrium with **lower price and quantity**, which on the diagram is shown at **point C**, left of the original equilibrium **E**.

10-C:

Answer: C (supply at different prices, assuming no changes in technology)

A **market supply curve** shows the quantity firms are willing to supply at various prices, **ceteris paribus** (holding technology, input costs, and other factors constant). This reflects the **law of supply**: as price rises, quantity supplied increases.

A: income levels relate to demand, not supply. And it does not affect the axes of the supply curve.

B is incorrect because factor prices (e.g. wages) affect **cost of production**, in turn causing **the supply curve to shift**, not the axes of a supply curve.

D refers to supply over time, which is not shown in the **static supply curve** in a price-quantity space.

11-B:

If someone loses their ability to drive, they will become more reliant on public transport, causing their **demand curve to shift rightward** from D1 to D2. This is a **non-price determinant of demand**, specifically, a **change in personal circumstances/preferences**. Hence, key B becomes correct.

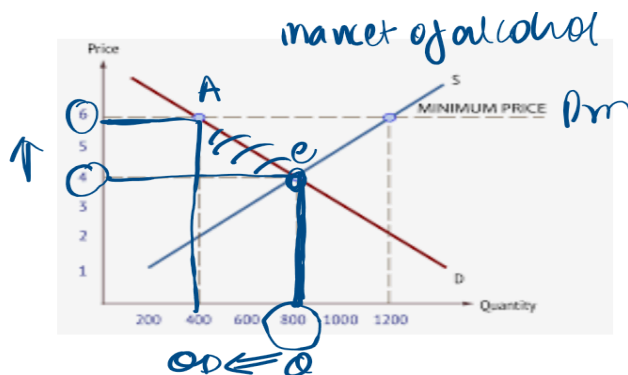
A is wrong: if car running costs fall, public transport becomes **less attractive**, shifting demand **left**, not right.

C is incorrect: a **fall in the price** of public transport causes a **movement along** the demand curve, not a shift.

D is wrong: reduced public transport services have to do with a decrease in supply.

12-A:

A) A minimum price set above the market equilibrium price effectively reduces the quantity demanded of the product. Governments often use this policy for demerit goods, which are considered harmful to consumers (like tobacco or alcohol). By increasing the price, the government discourages consumption of these goods.



B) Introducing minimum prices would make merit goods expensive, thereby reducing their consumption.

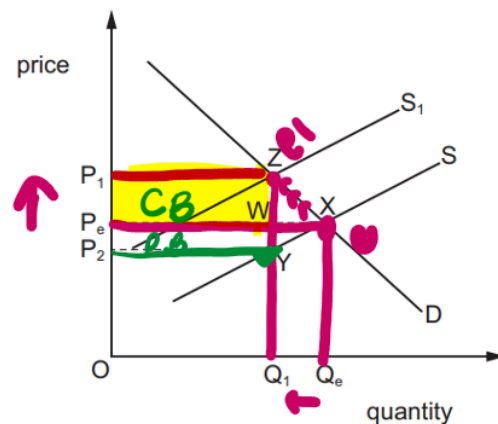
C) Minimum prices are aimed to artificially raise the prices of private goods.

D) An effective minimum price, which is set above the market equilibrium, leads to excess supply because producers are willing to supply more at the higher price, while consumers demand less. This imbalance results in a surplus of unsold goods. As a consequence, producer revenue may actually decline, since not all of their output can be sold at the new price. **Without additional government action, such as purchasing the surplus through buffer stock schemes, producer incomes are unlikely to improve and may even fall for those unable to sell their products.** Therefore, in the absence of further intervention, the policy fails to effectively support producer incomes, as clearly stated in the question.

13-B:

The **incidence of tax on consumers** is the difference between the **new price (P₁)** and the **original price (P_e)**, multiplied by the new quantity traded after taxation. This is shown through option B in the diagram.

13 The diagram shows the effect of an indirect tax imposed on cigarettes. The market is initially in equilibrium at point X.



Which area represents the incidence of the tax on consumers?

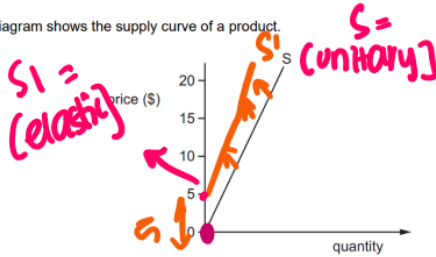
- A P_1ZYP_2 **B** P_1ZWP_e C P_eWYP_2 D P_eXZP_1

Option C (P_eWYP_2) represents the producer burden as this is the portion of the indirect taxation that can not be passed to consumers in form of a rise in prices. Hence, C will be incorrect.

14-D: The supply curve shown is a **straight line through the origin**, which has **unitary elasticity** ($p_{es}=1$) throughout.

Imposing a **specific tax** shifts the supply curve **upward (leftward) in parallel**, making supply more **responsive** at every point — hence **more elastic**.

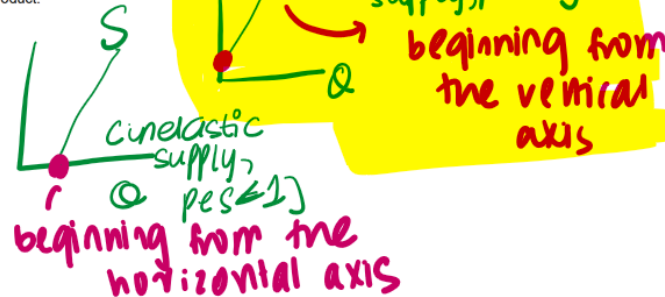
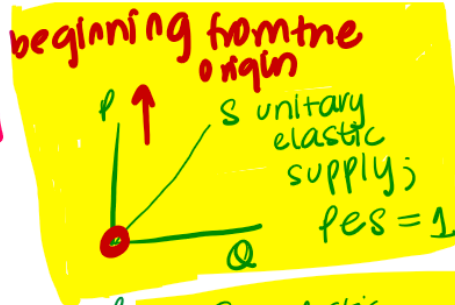
14 The diagram shows the supply curve of a product.



The government imposes a specific indirect tax of \$5 on the product.

How will the price elasticity of supply of the product change?

- A from elastic (>1) to inelastic (<1)
- B from inelastic (<1) to elastic (>1)
- C from inelastic (<1) to unitary ($=1$)
- ☒ D from unitary ($=1$) to elastic (>1)



15-A:

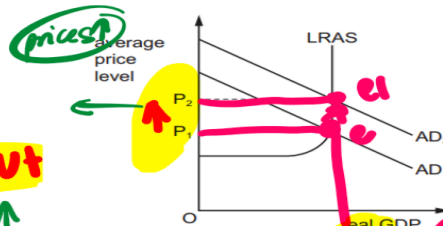
Reducing **labour costs** (a supply-side policy) reduces costs of production and shifts **aggregate supply rightward** to AS2. If **interest rates stay unchanged**, AD remains unchanged at AD1. The new equilibrium is at a **higher output level (real GDP)** and **lower price level** — shown in diagram A.

B and **D** are incorrect as they show **AD shifting**, which is not affected by labour cost changes. **C** is wrong because it shows a leftward shift in the AS curve.

16-A:

16 The diagram shows the effect on the average price level when aggregate demand (AD) increases from AD₁ to AD₂.

Handwritten notes on a yellow sticky note:
 nominal ↑
 • gdp ↑
 price ×
 real output
 • nominal gap ↑



Which statement relating to this change in aggregate demand is correct?

- ☒ A Nominal GDP has increased.
- B Nominal GDP is unchanged.
- ☒ C Real GDP has increased.
- ☒ D Real GDP has fallen.

Handwritten note on a yellow sticky note: 'if real GDP is unchanged'

Nominal GDP= GDP at current prices; value of output measured at current prices.

When aggregate demand increases from AD1 to AD2, the price level has increased from P1 to P2, and so, **the value of output measured at current prices has also increased. This is what we define as nominal GDP.**

Nominal GDP takes into account both changes in real output and the price level, so any increase in either component leads to an increase in nominal GDP.

17-C:

The economy described includes government spending and taxation, which are features of a government's involvement in the economy, indicating that it is not a purely market-based economy. Furthermore, the presence of imports and exports confirms that the economy is open to international trade. These two characteristics, government involvement and trade with other countries, are the defining features of **a mixed, open economy.**

For other options, see the image below:

17 The table lists the values of the components of an economy's circular flow of income.

Handwritten: Injections/leakages

	component	value \$m
<i>J</i>	government spending <i>G</i>	6
<i>J</i>	exports <i>X</i>	8
<i>J</i>	investment <i>I</i>	9
<i>J</i>	imports <i>M</i>	7
<i>W</i>	saving <i>S</i>	10
<i>W</i>	tax <i>T</i>	5

What can be concluded about the economy from the information shown?

- A It has a budget surplus. *X*
- B It has a trade deficit. *X*
- C* It is a mixed, open economy.
- D Its circular flow is in equilibrium. *X*

Handwritten: $J = W$ (equilibrium condition)

$$\begin{aligned} J &= 6 + 8 + 9 = 23 \\ W &= 7 + 10 + 5 = 22 \end{aligned} \quad \left. \begin{array}{l} \\ \end{array} \right\} J > W \Rightarrow \text{disequilibrium}$$

*Handwritten: A: $G > T \Rightarrow$ budget deficit
 $6 - 5 = \text{deficit of } \$1m$*

*Handwritten: B: $X > M = \text{trade surplus}$
 $8 - 7$
 $\hookrightarrow \text{surplus of } \$1m$*

18-C:

Negative economic growth implies a fall in real GDP, which usually leads to a decrease in the demand for labour. As firms cut back on production due to falling sales or output, they often reduce their workforce, leading to a rise in unemployment. This is a central concern during a recession and is strongly linked to the cyclical nature of unemployment in macroeconomic theory.

Option A is incorrect because a deteriorating balance of payments is not a result of negative growth. In fact, a recession can reduce imports and thus improve the current account.

Option B is incorrect since falling Aggregate demand typically leads to lower inflation or even deflation, not higher inflation.

Option D is incorrect because pollution often decreases when production levels fall, making it an unlikely cost of economic contraction.

19-C:

The description shows that the figure being calculated accounts for income from all sources (employment, pensions, benefits), **subtracts income tax and welfare contributions (making it disposable)**, and **adjusts for inflation (making it real)**. Therefore, this is a clear definition of real disposable income, which reflects the purchasing power households actually have to spend or save after taxation. Hence, C is correct.

Option A is incorrect because nominal income does not adjust for inflation.

Option B is wrong because although it considers net earnings (after tax), it does not account for inflation.

Option D is incorrect because gross earnings refer to income before tax, and again, this is not what the description represents.

20-C:

To prevent deflation, the government must increase aggregate demand.

Expansionary fiscal policy, such as increasing the budget deficit through higher government spending or lower taxes, stimulates demand directly.

Simultaneously, expansionary monetary policy, such as lowering interest rates, encourages borrowing and investment, thereby leading to a rise in aggregate demand.

Together, these policies work in the short run to increase consumption and investment, thereby raising the price level and preventing deflation in a closed economy where international influences are absent. Hence, key C becomes correct.

21-D:

Raising income tax reduces disposable income and therefore limits consumption. This helps control demand-pull inflation by reducing aggregate demand in the economy. Achieving price stability is a key macroeconomic objective, and taxation is often used as a fiscal tool to help moderate inflationary pressures.

Option A is incorrect because higher income tax does not depreciate the exchange rate. Option B is wrong as higher income tax may reduce economic growth by dampening aggregate demand. Option C is also incorrect because it may lead to higher demand-deficient unemployment due to lower demand for goods and services.

22-D:

Supply-side policies aim to improve the productive capacity and efficiency of the economy. This includes measures such as infrastructure investment, research and development, and education and training.

Option A is a valid supply-side policy because infrastructure development enhances long-run productive potential. Option B is also correct since research and development improve innovation and productivity. Option C is valid as subsidies for education and training improve labour quality and enhance efficiency.

Increasing the money supply (**option D**), however, is a monetary policy tool, not a supply-side policy, as it affects Aggregate demand in the economy. Hence, D will be the correct option.

23-B:

Answer: B (When it involves economy-wide institutions and behaviour)

Macroeconomics deals with the behaviour of the economy as a whole. This includes aggregate variables such as inflation, unemployment, GDP, and national income. A macroeconomic policy must therefore be one that affects or targets these broad aggregates or institutions that influence them, such as central banks or fiscal authorities.

Option A is incorrect because focusing on individual welfare is a feature of microeconomic policy. Option C is wrong as monopoly control is generally a microeconomic concern. Option D is also incorrect as buffer stock schemes are usually targeted at stabilising individual commodity markets, making them microeconomic in nature.

24-D:

->When the price elasticity of demand for exports is high, a depreciation in the exchange rate makes exports cheaper and more competitive abroad, leading to a proportionately greater increase in export volume. This will boost export revenues, helping to correct the deficit. Depreciation is also a classic expenditure switching policy tool, which helps to switch consumer demand from imported goods to domestically produced goods, again helping to reduce the deficit.

-> A rise in income tax, is a classic expenditure dampening policy tool, which will reduce the disposable incomes of individuals, thereby also leading to a reduction in the consumption of imported goods. This will reduce import expenditures, thereby helping to reduce the deficit.

Hence, key D will be correct.

25-B:

When the US restricts imports of cheap Chinese steel, domestic manufacturers who rely on low-cost steel face higher input costs. This reduces their competitiveness in global markets because they must now produce at higher costs compared to firms in other countries who still have access to cheap steel. This is a classic unintended consequence of protectionism, thus making key B correct.

Option A is incorrect because the price paid by US importers would rise due to restricted supply, not fall. Option C is not unintended — encouraging domestic steel production is often an intended goal of import restrictions. Option D is also an intended effect, aiming to support employment and incomes in protected industries.

26-B:

The current account records trade in goods and services, and tourism is a service export. When US tourists visit Jamaica and spend money on hotels, food, and entertainment, it results in an inflow of foreign currency, which is recorded as an export of services. Hence, B will be correct.

Option A is incorrect because purchasing shares is recorded in the financial account, not the current account. Option C is wrong as remittances sent abroad are part of secondary income, and outflows like these are recorded as debits. Option D is incorrect because it refers to a physical export of goods, not services.

27-A:

When a country with price-inelastic demand of gas imposes an import duty on gas, the price of gas rises significantly but the quantity demanded falls only slightly. This leads to a reduction in consumer surplus, the difference between what consumers are willing to pay and what they actually pay. Since consumers still need the good and have few substitutes, they continue buying at higher prices, suffering a welfare loss. This makes option A correct.

Option B is incorrect because the exporting country may lose revenue if demand contracts, and the benefit goes to the importing country's government who now receives revenue from taxes.

Option C is wrong as primary income relates to income from investments and employment, not import duties.

Option D is also incorrect as the duty imposed in one country does not raise the price for other importing countries.

28-D:

Countries specialise in producing the good for which they have a **comparative advantage**, meaning they can produce it at a **lower opportunity cost** than others. This allows them to use their resources more efficiently. When a country specialises, it **devotes all its resources** to producing just that one good, producing more than it needs.

The **surplus is exported** to other countries. In return, the country **imports the other good** from countries that specialise in it, creating mutual gains from trade. This way, both countries can enjoy a higher combined output than if they tried to produce both goods on their own.

28 The table shows the output of goods X and Y in China and the United States (US) before specialisation.

	good X	good Y
China	40,000	70,000
US	20,000	50,000
total	60,000	120,000

Assuming both China and the US use 50% of their resources to produce each product, what will the combined total output be after specialisation has occurred?

A 120,000 B 140,000 C 160,000 D 180,000

① calculate opportunity cost ratios -

② China = only produce Y now

A 50% resources allocated =
70,000 Y

* 100% resources allocated =
140,000 Y

B US = only produce X now

50% resources = 20,000 X; 100% resources allocated = 40,000 X

Total Output = 180,000

China
good X: good Y
20,000 X: 70,000 Y
1X: 3.5Y

Specialize in Y

US
good X: good Y
20,000 X: 50,000 Y
1X: 2.5Y

US has a lower opp cost when it produces X

Specialize in X

29-B:

Multilateral trade agreements (trade among more than 2 countries) allow countries to access larger markets, achieve economies of scale, and benefit from a more efficient global allocation of resources. In contrast, bilateral trade (trade among 2 countries) agreements are limited in scope and may prevent countries from tapping into potentially better trade deals with others. Multilateral trade promotes greater gains through more diversified and competitive exchanges. Hence key B is correct.

Option A is incorrect because trade diversion is not guaranteed to be greater in bilateral agreements. Option C is wrong as harm to third countries is not a guaranteed outcome of bilateral trade. Option D is incorrect because comparative advantage still exist in bilateral trade.

30-C:

Terms of trade (TOT) refers to the ratio of a country's export prices to its import prices. It indicates how much of a country's exports are needed to obtain a given quantity of its imports. Essentially, it reflects the purchasing power of a country's exports in terms of its imports.

Its formula is:

$(\text{Index of export prices} / \text{Index of import prices}) * 100$

For Terms of Trade to improve, export prices should increase while import prices should decrease.

A rise in the exchange rate of a country or appreciation leads to a rise in export prices of a country relative to its import prices which fall. This will result in an improvement in the TOT, hence making option C correct.

Option B, a relatively low rate of domestic inflation, would result in relatively lower export prices. Hence, TOT is likely to worsen.

Changes in budget deficit or changes in the level of primary income do not directly affect a country's TOT, hence making options A and D incorrect.

